

Multiple Choice

1. **Answer:** A

Options: A) Am; B) As; C) Ab; D) Au; E) Fe

Note: The correct symbol for antimony is "Sb," not "Am," as "Sb" is derived from the Latin name "stibium," which is the accepted chemical symbol in the periodic table for this element.

2. **Answer:** D

Options: A) 2.11×10^{-8} ; B) 2.73×10^{-8} ; C) 1.08×10^{-8} ; D) 3.92×10^{-8} ; E) 4.77×10^{-8}

Note: The correct answer of 3.92×10^{-8} for the K_{sp} of CaF_2 is derived from the solubility concentration and F^-) based on the stoichiometry of the dissolution reaction.

2. **Answer:** C

Options: A) $1s < 2p < 2s < 3p < 3s < 3d$; B) $2s < 2p < 1s < 3s < 3p < 3d$; C) $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < 5s = 5p = 5d = 5f = 5g$; D) $2s = 2p < 1s = 3p = 3d$; E) $1s < 3s < 2s < 5s < 4s$

Note: The correct answer reflects the increasing energy levels of atomic orbitals as dictated by quantum mechanics, where the energy of orbitals increases with their principal quantum number and subshell type, leading to the specified order of $1s$, $2s$, $2p$, and so forth, with the added consideration of electron-electron interactions in multi-electron atoms.

3. **Answer:** C

Options: A) 1.25×10^3 K; B) 1.20×10^3 K; C) 1.41×10^3 K; D) 1.10×10^3 K; E) 1.65×10^3 K

Note: The correct answer of 1.41×10^3 K is derived from the van der Waals equation, which incorporates the specific parameters of chlorine to determine the temperature at which the gas behaves ideally, indicating the Boyle temperature.

4. **Answer:** B

Options: A) Al; B) Tl; C) C; D) B; E) Si

Note: The +1 oxidation state is more stable than the +3 oxidation state for thallium (Tl) because of the inert pair effect, where the s-electrons are less available for bonding, leading to a preference for the lower oxidation state in heavier group 13 elements.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the solutions to the equation $x^2 - 5x + 6 = 0$,

when correctly calculated using the quadratic formula, are $x = 2$ and $x = 3$, not $x = 4$ and $x = 5$.

7. **Answer:** True

Options: A) True; B) False

Note: Methylamine can form hydrogen bonds because the nitrogen atom possesses a lone pair of electrons that can interact with hydrogen atoms of other molecules, allowing for the formation of these attractive interactions.

8. **Answer:** True

Options: A) True; B) False

Note: The ^{13}C spectrum of 3-ethylpentane exhibits five distinct chemical shifts because the unique arrangement of its carbon atoms leads to different electronic environments, resulting in five non-equivalent carbon signals.

9. **Answer:** True

Options: A) True; B) False

Note: Covalent network bonding results in high melting points due to the strong, extensive networks of covalent bonds that require significant energy to break, while poor electrical conductivity arises because these substances lack free-moving charged particles to carry current.

10. **Answer:** False

Options: A) True; B) False

Note: The logarithm of 3^2 is equal to 2, not 3, because the logarithm function represents the exponent to which the base must be raised to yield that number, so $\log(3^2) = 2$.

Long Form Response

11. **Answer:** The chemical shift in NMR spectroscopy is influenced by several factors, including electronegativity, hybridization, molecular symmetry, and the presence of nearby functional groups. In the case of CH 3 Cl, the chlorine atom is highly electronegative, which induces a downfield shift of the hydrogen resonances due to deshielding effects. The ^1H resonance of CH 3 Cl appears approximately 1.55 kHz from TMS in a 12.0 T magnetic field because the strong magnetic field enhances the effects of the induced electron density changes around the hydrogen atoms, leading to a significant chemical shift. Additionally, the molecular geometry and the dipole moment of CH 3 Cl contribute to the specific resonance frequency observed in the NMR spectrum.

Note: correct because it identifies the key factors influencing chemical shifts, such as electronegativity and deshielding effects from the electronegative chlorine atom in CH 3 Cl, which contribute to its specific resonance frequency in a strong magnetic field.

12. **Answer:** Cobalt-60 is produced from cobalt-59 through a nuclear reaction known as neutron bombardment, where cobalt-59 nuclei capture neutrons. This process typically occurs in a nuclear reactor, where neutrons are abundant; cobalt-59 absorbs a neutron to become cobalt-60, which is radioactive and decays by emitting beta particles and

gamma rays. The emitted gamma rays are particularly useful in radiation therapy, as they can penetrate tissues and target cancer cells effectively while minimizing damage to surrounding healthy tissue. This transformation underscores the importance of neutrons in nuclear chemistry and their practical applications in medical treatment, particularly in the field of oncology.

Note: correct because it accurately describes the nuclear reaction in which cobalt-59 captures a neutron to become cobalt-60, and it highlights the significance of the resulting gamma radiation in effectively targeting cancer cells during radiation therapy.

End of Answer Key